

How Many Retail Trading Strategies Survive Honest Evaluation? A Systematic Falsification Study

Anupam Patil · Working paper, July 2026 · Draft v0.2 (assembled from the project repositories; all numbers machine-generated by the cited code)

Code and data: github.com/jarvisss007 — `backtest-overfitting` (toolkit), `strategy-lab` (survey), `dc-ml-trading`, `regime-monte-carlo` (companion papers).

Abstract

Retail trading culture supplies an endless stream of strategies that look profitable in backtests. Academic finance has documented the mechanism — multiple testing, backtest overfitting, and execution-assumption errors inflate in-sample performance (Bailey & López de Prado 2014; Bailey, Borwein, López de Prado & Zhu 2015) — but retail evaluations rarely apply the corrections. This paper applies a uniform "honest evaluation" gate (Deflated Sharpe Ratio, Probability of Backtest Overfitting via CSCV, minimum backtest length, cost-aware walk-forward, and strict execution realism) to a broad sample of strategies a retail trader would actually consider: nine classic equity strategy families over 15 years; a popular commercial TradingView "Smart Money Concepts" system; intraday timing, sentiment, mean-reversion, and event studies on S&P data; cryptocurrency order-flow signals; two folklore claims taken directly from best-selling price-action trading books; and fifteen years of Indian index weekly-options structures. The result is nearly uniform: **no market-neutral or options strategy tested survives the gate**. The only survivors are long-only variants of risk-managed equity beta, which add no deflation-robust alpha over buy-and-hold. Several strategies fail in instructive ways: a lookahead bug masquerading as edge (PBO 0.60), an edge that exists only in an untradeable execution window, statistically significant microstructure signals that are net-negative after costs, premium-selling structures that win up to 86% of weeks while compounding to losses, and a near-universally taught "wait for confluence" rule that degrades out-of-sample performance rather than improving it. We argue the honest-evaluation gate is cheap, reusable, and should be the default before any capital — and we document a pre-registered forward-testing protocol now running as the out-of-sample continuation of this study.

1. Introduction

The replication crisis in factor research (Harvey, Liu & Zhu 2016; Hou, Xue & Zhang 2020) has a retail sibling that is larger and almost entirely unexamined. Strategy ideas reach retail traders through backtest screenshots — from social media, best-selling trading books, strategy marketplaces, and charting-platform script libraries — evaluated without multiple-testing corrections, without out-of-sample discipline, and frequently with subtle lookahead or execution errors. The tools to audit these claims have existed for a decade, but they live in institutional quant literature, not where retail decisions

are made.

This paper does three things. First, it packages the audit machinery as a small open-source toolkit and validates it on synthetic data (§2). Second, it applies one uniform gate to a deliberately heterogeneous sample of strategies spanning what a motivated retail trader in 2026 actually encounters (§4–§8). Third, because even honest backtests remain vulnerable to selection at the researcher level, it registers a live forward-testing protocol whose predictions are timestamped, scored without discretion, and publicly aggregated (§9).

The contribution is not a new anomaly — it is the opposite: a documented, reproducible demonstration of how many popular strategies fail when evaluation is honest, and a catalogue of the specific failure modes that make them look like they work.

2. The honest-evaluation gate

A strategy claim passes the gate only if all of the following hold:

1. **Deflated significance.** The Deflated Sharpe Ratio (DSR; Bailey & López de Prado 2014) corrects the observed Sharpe for the number of trials, non-normality (skew, kurtosis), and sample length, benchmarked against the expected maximum Sharpe of equally-many noise strategies. We require $DSR \geq 0.95$.
2. **Low probability of backtest overfitting.** PBO estimated by combinatorially symmetric cross-validation (CSCV; Bailey, Borwein, López de Prado & Zhu 2015). We flag $PBO > 0.10$ and report the in-sample→out-of-sample rank degradation slope.
3. **Sufficient sample.** Minimum backtest length (MinBTL) given the trial count; e.g., selecting among 1,000 configurations at target Sharpe 1.0 requires roughly a decade of data before the winner means anything.
4. **Execution realism.** Next-bar-open fills, stop-loss-priority on ambiguous bars, completed-bar (non-repainting) indicator values, published-data lags (e.g., sentiment used only after its release date), and explicit transaction costs.
5. **Untouched holdout.** Where a parameter grid was searched, a 70/30 temporal holdout the search never saw.

Toolkit validation. On 200 pure-noise strategies the best in-sample Sharpe is 1.34 — and the gate rejects it (DSR 0.36, PBO 0.50). A planted Sharpe $\approx$ 3 edge among 199 noise strategies passes (DSR 0.998, PBO 0.04). The instrument distinguishes signal from selection.

3. Study 1 — Systematic survey of classic strategy families

Nine strategy families (eight substantive plus a day-of-week noise control), full parameter grids, 15 years of daily data (2011-07 → 2026-07), 119-ticker universe plus SPY, 5 bps per unit turnover, one-day trade lag. Benchmark: SPY buy-and-hold Sharpe 0.921 over the same window.

Family	Configs	IS Sharpe	DSR	PBO	OOS Sharpe	Verdict
MA trend basket	8	1.20	1.00	0.12	1.47	survives*
Turn-of-month	3	1.09	1.00	0.08	1.21	survives*
Vol-managed SPY	6	0.99	1.00	0.37	0.95	survives*
Donchian breakout	6	1.19	1.00	0.57	1.35	suspect
52-week high/low	17	1.15	0.48	0.06	1.21	suspect
TS momentum	24	1.32	0.94	0.07	1.10	suspect
XS momentum	36	0.50	0.75	0.21	0.20	suspect
Short-term reversal	18	0.05	0.00	0.47	-0.74	suspect
Day-of-week (control)	5	-0.06	0.17	0.68	-0.56	overfit ✓control

The asterisk is the finding. Every formal survivor is a long-only construction whose returns are dominated by equity beta over a window whose out-of-sample segment ($\approx 2022-2026$) contains exactly one bear market. None demonstrates deflation-robust alpha over simply holding the index; the universe itself (today's watchlist) embeds survivorship bias, and selecting the best family across nine adds a layer of multiple testing the per-family gate does not count. The noise control behaving as noise is the quality check that the gate is not vacuously permissive.

Conclusion of Study 1: zero market-neutral edges; "what survives" is risk-managed beta.

4. Study 2 — Anatomy of a commercial strategy: the r17 conviction

"IE Pro r17" is a commercial TradingView Smart-Money-Concepts system (EMA/VWAP/RSI scoring plus liquidity sweeps, break-of-structure, fair-value gaps, order blocks). We ported it 1:1 to Python with honest fills and non-repainting indicator values, then evaluated it four independent ways:

1. **Causal replay.** The TradingView backtest's profitability reproduces only when the platform's `request.security` lookahead behavior (seeing up to 14 minutes ahead) is reproduced. Causal evaluation collapses profit factor from 1.44 to 0.57 (short-only, pseudo-OHLC); raw sweeps on true OHLC test at $t = -2.9$. **The backtest performance was the bug.**
2. **Deflation of the audit sweep.** A 6-configuration audit of the oDTE variant: the in-sample winner was exactly the configuration with the lookahead error; DSR 0.34, PBO 0.60, IS→OOS slope $-1.44 \rightarrow$ overfit.
3. **Variant sweep.** 11 strategy variants including inversions: best (fading r17) posts annualized Sharpe 3.24 — against an expected best-of-11-noise Sharpe of 3.62; DSR 0.42 $\rightarrow$ unproven.
4. **Condition and ML referendum.** 38 entry conditions under triple-barrier labeling with Bonferroni correction: zero survivors on both 5-minute and hourly data. The single provisional survivor (near-prior-day-high, 64.9% up over $n=131$) was taken to a 7-year referendum: 779 events, 52.5% up ($z = 0.88$), net $-0.225R$ ($t = -6.2$) — convicted. A LightGBM/logistic walk-

forward layer on the full feature set: out-of-sample AUC within 2 standard errors of 0.5 — a coin flip.

One commercial system, four independent methods, one answer.

5. Study 3 — Breadth: the same gate across asset classes

- **Intraday timing (S&P, 302 sessions of minute data).** The "session timing" folklore reduces to the long-documented U-shape: moves cluster at the open and close (up to 21.5% vs 10% baseline frequency, Bonferroni-significant) — a description of variance, not an edge; a 17-timestamp $\times$ 4-holding grid produces a best t of 2.68 against an expected max-noise t of 2.39, with zero Bonferroni survivors.
- **Daily mean-reversion (32 years).** The pattern is statistically real (all nine down-streak longs positive; PBO 0.002) but fails deflation (DSR 0.59), and its adaptive walk-forward implementation earns +2.6%/yr (Sharpe 0.26) against buy-and-hold's +10.2%/yr (Sharpe 0.53). *Real anomaly, no retail edge* — a category this literature needs.
- **Sentiment timing (AAII, 39 years, publication-lagged).** Best of 24 variants is "always long" (Sharpe 0.64 = the index); DSR 0.23; the flagship conditional strategy underperforms holding SPY on its own selected days.
- **Insider-cluster gap (SEC Form 4).** Entry at the filing-day close yields net Sharpe +3.1 — but that close is not purchasable when filings arrive after hours. At the honest next-open entry, every variant is net-negative (all $|t| < 2$). **The edge lives entirely inside an untradeable execution window** — the cleanest demonstration we have that execution assumptions are a multiple-testing dimension of their own.
- **Crypto order flow (Coinbase, 187k trades).** Order-flow imbalance predicts 5–60s returns with information coefficients of 0.12→0.03, permutation $p = 0.002$ at every horizon — genuinely significant — and every configuration is net –60 bps per trade after realistic costs (hit rates 12–23%). Statistical significance and money are different quantities.
- **ML on directional-change events (companion paper).** Gradient-boosted filters on DC events across S&P, NASDAQ, WTI (1986–2019): no bootstrap p -value below 0.05; the draft literature's headline "+37.8%" WTI result lands at +1.4% under independent reimplementations.
- **Event studies (war escalation/de-escalation, 24 events).** De-escalation drift of +142→+240 bps (bootstrap $p \approx 0.014$ –0.024 at 1d/5d) is *suggestive, not conclusive* at $N = 8$, and is reported as such rather than traded.

6. Study 4 — Fifteen years of index weekly-options structures

Using 783 weekly cycles of real BankNifty closes and real India VIX (2011–2026), with synthetic flat-vol Black-Scholes premiums (India VIX $\times$ 1.25, 6.5% rate, 1.5%/leg/side costs):

Structure	Win rate	Profit factor	Ann. return	Ann. Sharpe
Short 1 σ strangle	73.6%	0.67	–6.4%	–0.83

Structure	Win rate	Profit factor	Ann. return	Ann. Sharpe
Iron condor	71.0%	0.76	−2.6%	−0.74
Short ATM straddle	55.6%	0.72	−10.0%	−0.84
Put credit spread	86.1%	0.89	−0.6%	−0.24
Long ATM straddle	42.0%	1.15	+4.4%	+0.37
Long index (no options)	53.9%	1.31	+13.5%	+0.70

Every premium-selling structure loses after costs despite win rates up to 86% — the high-win-rate/negative-expectancy trap measured at scale, with settlement computed on real index paths. The best performer in the tournament is *not trading options at all*, and even it fails deflation once the six trials are counted (DSR 0.256; verdict: unproven). The lone positive options result (long straddles) sits precisely on the model's weakest assumption — flat volatility omits the smile premium real buyers pay — and is therefore recorded as a hypothesis for real-chain data, not a finding. Caveats cut both ways and are documented in the repository.

7. Study 5 — Trading-book folklore: does "confluence" beat a single signal?

Books are the oldest retail strategy channel and the most authority-coded — a reader trusts a paperback differently than a forum screenshot. Two claims from popular price-action titles (*Price Action Breakdown*; *The Ultimate Guide to Price Action Trading*) are concrete enough to run on the Study 1 apparatus: 15 years of daily data, 119 names plus SPY, 5 bps per unit turnover, one-day lag.

Claim (i) — value-area reversion. "Price reverts toward its value area." Operationalized as a cross-sectional fade of each name's distance from its 20-day mean (long the most-below, short the most-above), across six configurations (fade quantile $\in \{0.1, 0.2\} \times$ holding $\in \{5, 10, 21\}$ days). The best config (q0.2, 21-day hold) posts in-sample net Sharpe 0.08 and walk-forward 4/5 — but out-of-sample net Sharpe −0.03 and DSR 0.16. The direction is real; the edge is not. Verdict: *real-but-loses-to-buy-and-hold* — the same category as the daily mean-reversion result of §5.

Claim (ii) — confluence. "Never trade a single signal; wait for confluence." This is arguably the single most repeated instruction in retail technical education, and it is directly testable: take the identical value-area fade, but count a position only where a longer-horizon trend *agrees* with it. If confluence is the edge the books claim, the gate should improve the fade. It does the reverse — every confluence configuration underperforms the ungated fade, walk-forward scores collapsing to 0–1 of 5. The confirming condition added no signal; it subtracted sample and supplied a narrative. Verdict: **dead — refuted.**

The second result is the transferable one. "Confluence" feels like discipline — more conditions, more confirmation, more patience — but in this universe the extra condition is a selection knob wearing the costume of prudence, and it makes the strategy strictly worse. We record it as a sixth failure mode, *complexity comfort*, in §8.

8. What survives, and the taxonomy of false edges

Across every study, the surviving set is: **long-only equity exposure and its risk-managed variants** — none of which demonstrates deflation-robust alpha over the index itself. The false edges, meanwhile, fall into six reproducible categories:

1. **Selection** — the best of N trials reported as if N = 1 (r17 variants, timestamp grids, sentiment variants).
2. **Lookahead** — indicator or data timing that quietly sees the future (the r17/request.security collapse; rebuilt forward-return labels that shrank a "+0.169% edge" to +0.009%).
3. **Execution fiction** — fills at prices unavailable in practice (the insider-gap conviction).
4. **Cost blindness** — significant signals that are net-negative after realistic frictions (crypto OFI; oDTE momentum at an 11% pre-cost win rate).
5. **Distributional camouflage** — high win rates concealing negative expectancy (premium selling, §6).
6. **Complexity comfort** — confirming conditions layered on for the feeling of rigor ("wait for confluence") that add no out-of-sample signal, and here subtract it (the value-area confluence gate, §7).

9. The forward-test protocol (pre-registered, running)

Backtests — even honest ones — leave residual researcher degrees of freedom. Since July 2026, five live "agents" (US equities, Indian indices, insider clusters, oDTE chains, crypto order flow) log falsifiable, timestamped predictions to append-only ledgers, each scored mechanically on its check date under a written no-discretion rule, with results pooled in a public scoreboard. Hit rates near coin-flip will be reported as findings, consistent with the rest of this paper. This converts the study from a retrospective audit into an ongoing pre-registered experiment.

10. Limitations

Single researcher; free data (Yahoo, EDGAR, exchange feeds) with their known defects; the survey universe embeds survivorship bias (stated where it binds); the options study uses synthetic premiums pending real chain data; several sub-studies have modest N (war events); the trading-book claims are tested only on this US large-cap daily cross-section, not the intraday charts their authors typically use, so their failure here refutes the stated logic in this universe rather than everywhere; and cross-study selection — the choice of which strategies to test — is itself a bias this paper cannot fully escape, only disclose.

11. Conclusion

Given one uniform, honest gate, the retail strategy landscape thins to almost nothing: beta survives, alpha claims do not. The practical implication is not nihilism — it is sequencing. The gate costs an afternoon; the strategies it rejects cost accounts. Evaluation this cheap should precede capital every

time.

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Draft assembled 2026-07-15. v0.2 (2026-07-18) adds Study 5 (trading-book folklore — the value-area fade and the "confluence" refutation, from the Fluency-Project book-cluster experiments logged in `discoveries.csv` on 2026-07-11) and the sixth taxonomy category, "complexity comfort." TODO before any submission/posting: (1) re-verify every number against a fresh run of each repository's pipeline; (2) decide SSRN/arXiv posting (currently shelved per standing decision); (3) add exhibits (deflation curves, the CSCV rank plots, the options P&L distributions); (4) replace the \$9 placeholder stats with the first 100+ scored forward calls.